

AKRON FULTON, OH, USA

WMO: 724303

Lat: 41.038N

Lon: 81.464W

Elev: 318

StdP: 97.56

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 14813

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.0	-13.1	-20.4	0.6	-15.0	-17.8	0.8	-12.1	11.4	0.4	10.5	-1.5	3.1	250	0.494

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.4	32.0	22.5	30.7	22.0	29.3	21.3	24.1	29.4	23.3	28.3	22.6	27.3	4.4	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.4	17.8	27.0	21.8	17.1	26.2	21.1	16.3	25.5	73.8	29.9	70.7	28.2	67.8	27.2	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.5	8.4	7.6	-19.1	33.7	3.9	1.5	-21.9	34.8	-24.1	35.7	-26.3	36.6	-29.1	37.7		
(5)			-19.5	25.5	3.7	0.8	-22.2	26.0	-24.3	26.4	-26.4	26.9	-29.0	27.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.8	-2.7	-1.4	3.5	10.3	16.2	20.7	23.0	22.3	18.6	12.0	5.7	0.4		
	DBStd	10.37	6.61	6.32	6.23	5.33	4.84	3.62	2.98	2.91	4.11	4.97	5.09	5.66		
	HDD10.0	1489	394	324	221	60	6	0	0	0	1	35	147	300		
	HDD18.3	3221	651	554	461	244	100	19	2	4	46	206	379	555		
	CDD10.0	1776	2	4	18	71	199	322	402	380	259	96	18	3		
	CDD18.3	466	0	0	1	4	35	91	146	126	54	9	0	0		
	CDH23.3	3834	0	0	5	51	314	736	1244	986	439	58	1	0		
	CDH26.7	1139	0	0	0	4	70	213	414	295	131	11	0	0		
(14) Wind	WSAvg	3.3	4.2	3.9	3.8	3.7	3.0	3.0	2.6	2.4	2.5	3.1	3.5	3.9		
(15) Precipitation	PrecAvg	996	72	54	71	87	101	90	112	98	92	79	70	72		
	PrecMax	1305	149	125	146	146	199	149	290	205	182	156	143	175		
	PrecMin	651	35	26	13	41	40	13	31	27	31	27	36	23		
	PrecStd	150	29	25	30	27	46	39	64	41	40	34	26	31		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.2	19.7	23.4	27.4	30.5	32.4	33.7	33.1	32.5	28.0	22.0	17.2		
		MCWB	12.8	13.9	15.4	17.4	21.1	22.5	24.0	22.8	22.2	20.8	15.6	13.2		
	2%	DB	13.4	14.6	19.4	24.8	28.6	30.9	32.1	31.3	30.3	24.8	18.6	13.9		
		MCWB	11.3	11.1	13.3	15.9	19.9	21.8	23.0	22.2	21.6	18.2	13.8	10.9		
	5%	DB	10.2	11.3	16.4	22.4	27.0	29.2	30.6	29.8	27.8	22.5	16.3	11.6		
		MCWB	7.9	8.3	11.6	14.6	19.0	21.2	22.1	21.6	20.3	17.3	12.0	9.2		
	10%	DB	7.1	7.8	13.8	19.9	24.9	27.5	29.0	28.1	25.9	20.2	13.9	8.7		
		MCWB	4.8	4.8	9.7	13.1	17.9	20.1	21.3	20.9	19.4	15.8	10.4	6.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.8	14.6	16.1	19.2	22.8	24.1	25.6	24.9	24.1	21.8	17.2	14.2		
		MCDB	15.5	18.3	21.7	25.1	29.0	30.2	31.4	30.5	30.2	27.0	20.1	16.0		
	2%	WB	11.6	11.6	14.4	17.2	21.2	23.0	24.3	23.8	22.8	20.0	14.8	11.7		
		MCDB	13.5	13.9	18.4	22.4	26.7	28.6	29.6	28.5	27.8	23.5	17.9	13.4		
	5%	WB	8.2	8.2	12.4	15.7	20.1	22.1	23.5	22.9	21.7	18.0	12.7	9.2		
		MCDB	9.8	11.4	15.7	20.9	25.2	27.4	28.4	27.2	25.8	21.3	15.5	11.0		
	10%	WB	4.8	4.9	10.0	14.2	18.9	21.3	22.6	22.1	20.5	16.2	10.8	6.7		
		MCDB	6.9	7.8	13.6	18.8	23.6	26.0	27.2	26.3	24.3	19.6	13.6	8.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.6	8.6	9.7	11.3	11.2	10.4	10.4	10.3	10.7	9.8	8.6	6.9		
		MCDBR	10.5	12.6	13.6	14.6	13.1	12.0	11.5	11.8	12.5	12.5	12.4	10.3		
	5% WB	MCWBR	9.2	9.9	8.8	8.5	6.6	5.6	4.7	4.8	5.5	7.1	8.5	8.7		
		MCWBR	10.0	11.7	12.2	12.4	11.3	10.4	9.7	9.4	10.0	10.3	10.8	9.9		
(40) Clear-Sky Solar Irradiance	taub		0.301	0.329	0.356	0.389	0.427	0.449	0.450	0.439	0.384	0.362	0.319	0.303		
	taud		2.425	2.330	2.338	2.281	2.200	2.178	2.193	2.215	2.343	2.398	2.493	2.462		
	Ebn at Noon		855	877	892	883	852	832	826	825	851	829	830	823		
	Edh at Noon		79	101	113	128	143	146	143	136	111	94	74	71		
(44) All-Sky Solar Radiation	RadAvg		1.43	2.30	3.28	4.48	5.36	5.88	5.88	5.29	4.24	2.76	1.78	1.18		
	RadStd		0.12	0.29	0.33	0.43	0.46	0.45	0.40	0.27	0.39	0.25	0.27	0.12		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+1.30	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+269	+182
(47)	Regional (50 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+112	+60

Nomenclature: See separate page